Unsafe Outdoors:

An Analysis of Extreme Temperature and Ozone Conditions in Louisville to Support Emergency Shelter Planning

Why is immediate shelter needed?

- Weather exposure is a **public health emergency** in Louisville.
- Unsheltered individuals are exposed to dangerous temperatures and air quality **year-round**.
- For those without shelter, the weather is not an inconvenience—it’s **life-threatening**.

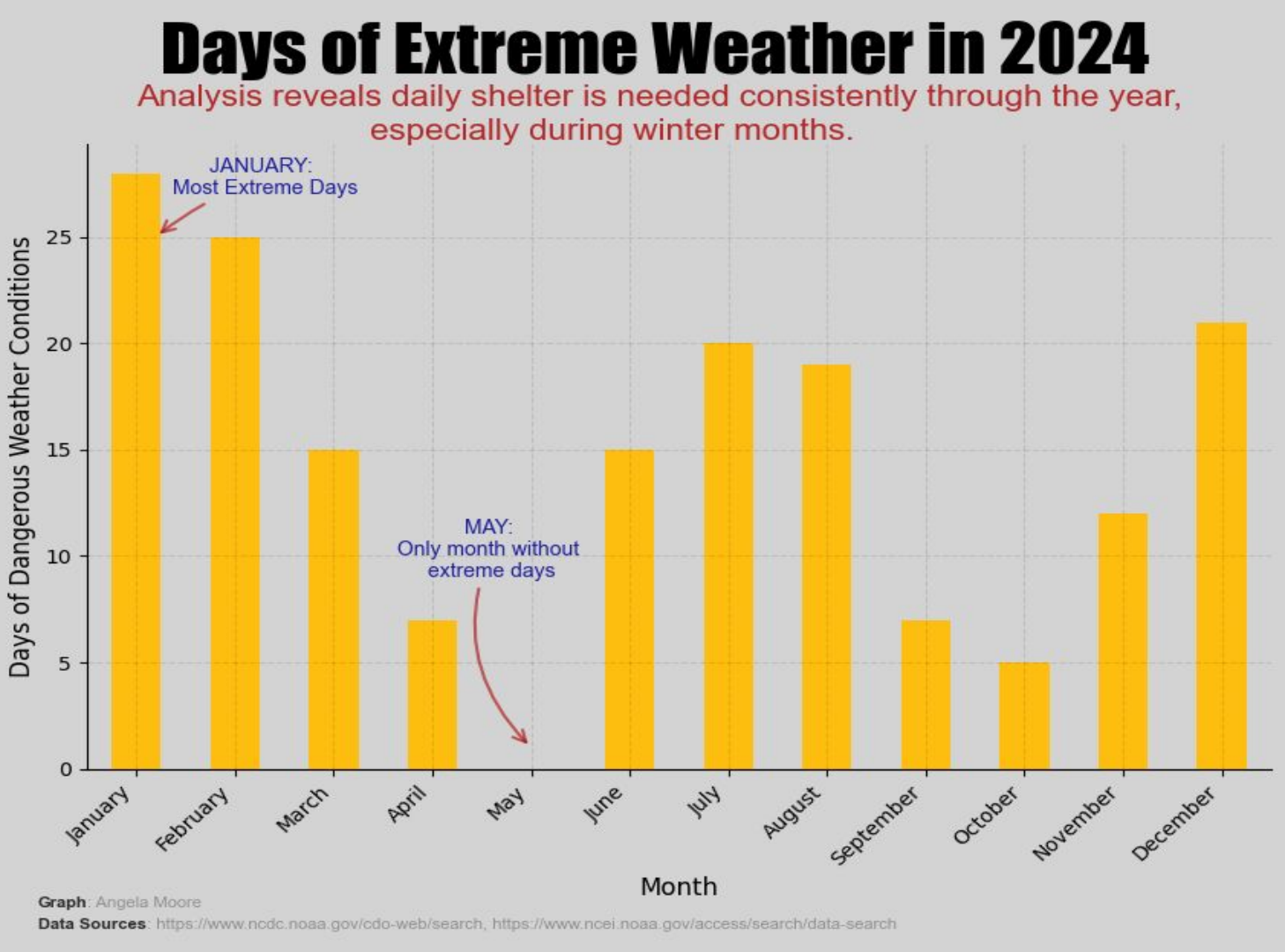
What does dangerous weather look like?

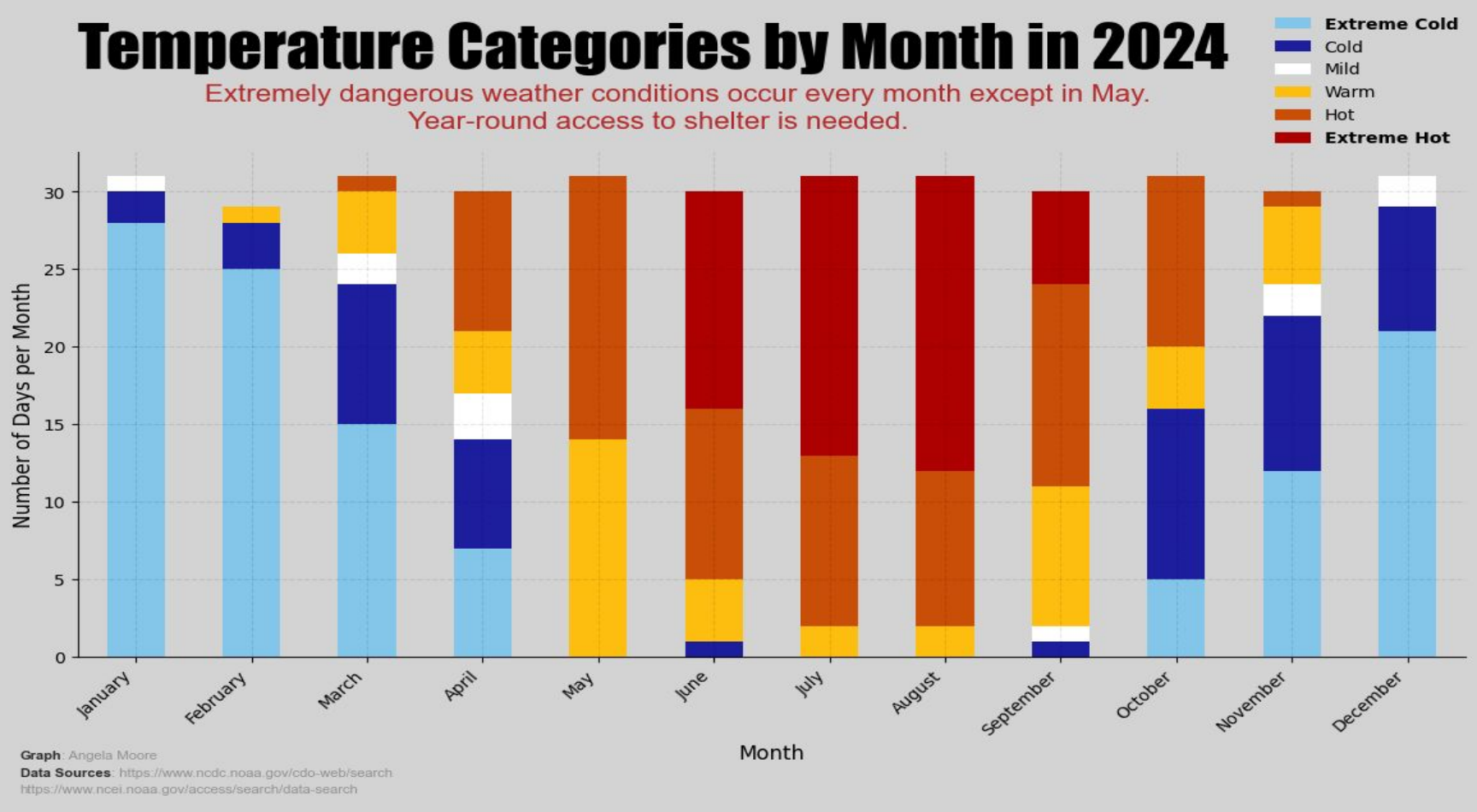
- Heat (Heat index at or above 95°F)
- Cold (Windchill at or below 35°F)
- Air Quality (Ozone AQI levels of 100 and above)

Why does *immediate* shelter matter?

- Exposure happens DAILY, not seasonally.

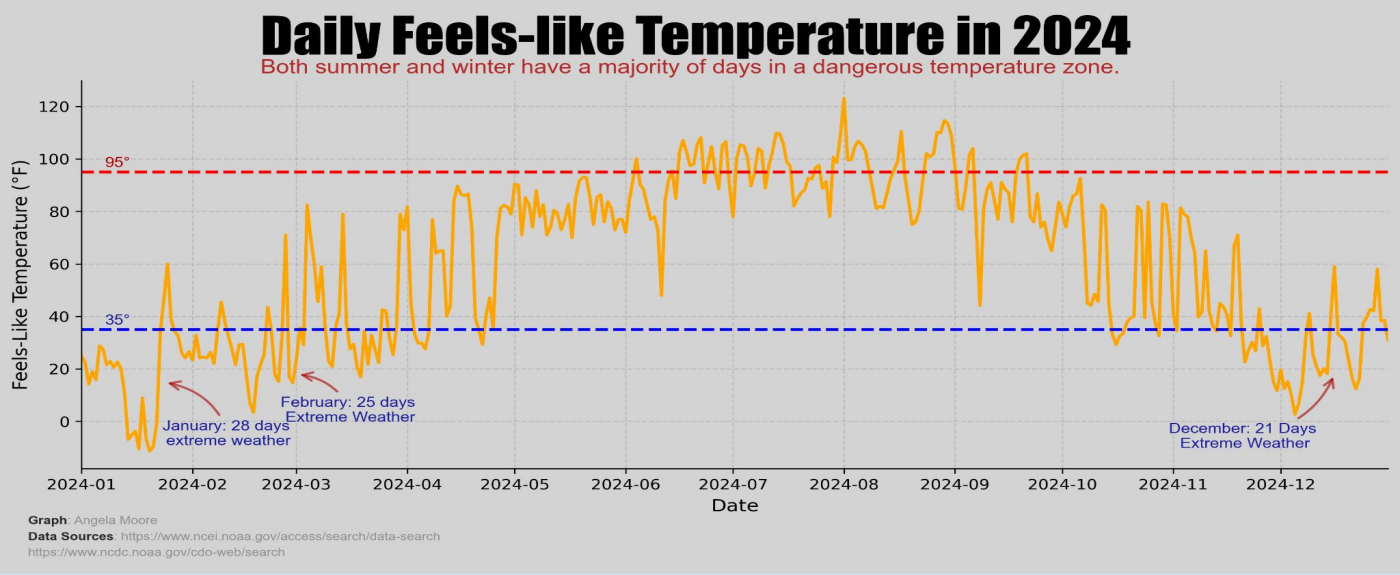
- ➔ Dangerous conditions occur consistently through the year.
- ➔ Shelter is needed beyond the traditional “winter-only” models.

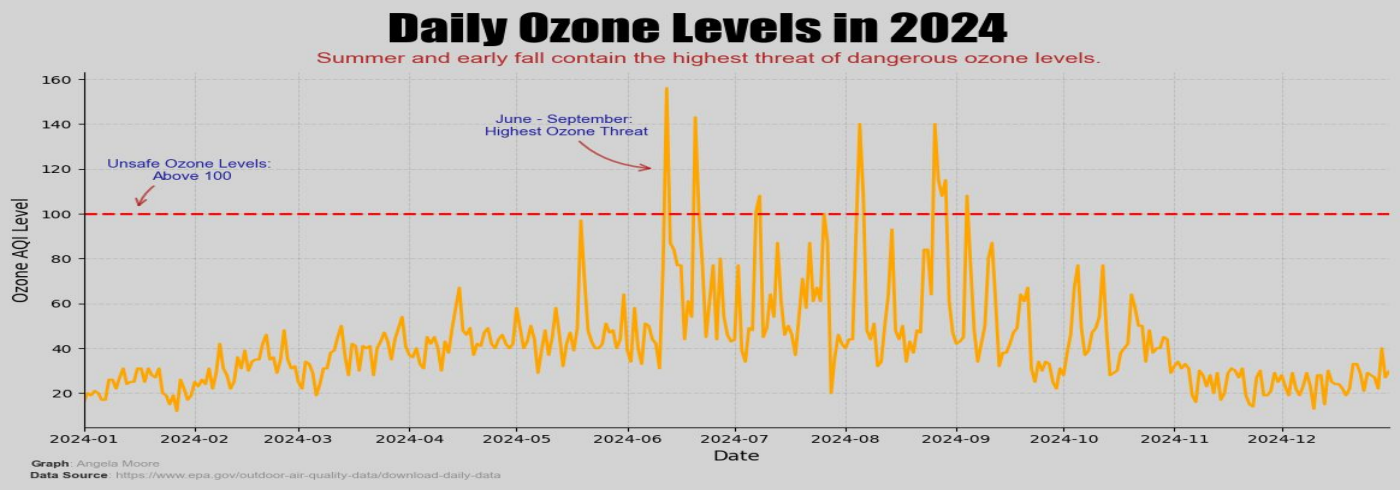


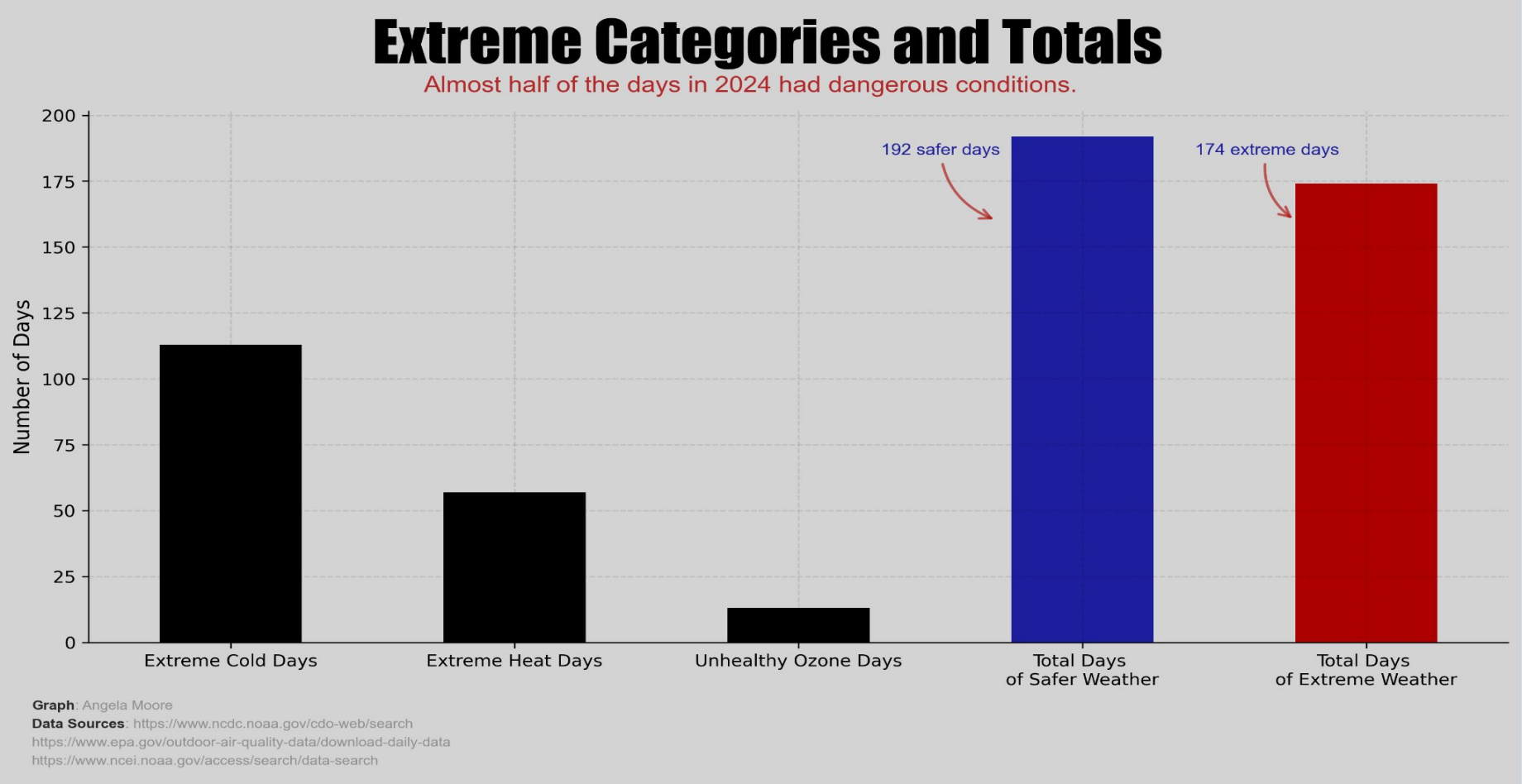


Even when conditions don’t meet “extreme” thresholds, many additional days still force unsheltered people to endure ongoing discomfort with no relief.

- Though winter has the most days of extreme temperature conditions, also looking at unsafe air quality levels show that summer and fall are consistently dangerous as well.







These numbers capture only the most dangerous days, not even considering the far greater number of days that are still uncomfortably harsh for unsheltered individuals.

What does this mean for someone without shelter?

- No escape from heat or cold
- Continuous breathing of polluted air
- Increased ER visits, illness, and mortality risk

Indoor refuge protects most residents. Without shelter, extreme weather becomes a ***life-threatening*** exposure.

Responding to this Immediate Risk:

- Treat shelter as a **critical health intervention**, not just housing support
- Integrate weather and ozone data into **citywide emergency response planning**
- Develop **low-barrier, rapid-access shelters** during extreme conditions
- Open **temporary or pop-up shelters** during extreme weather
- Expand **transportation options** to shelters during extreme weather events
- Shift from reactive responses to **proactive, climate informed shelter planning**

Weather does not wait — and neither can we.